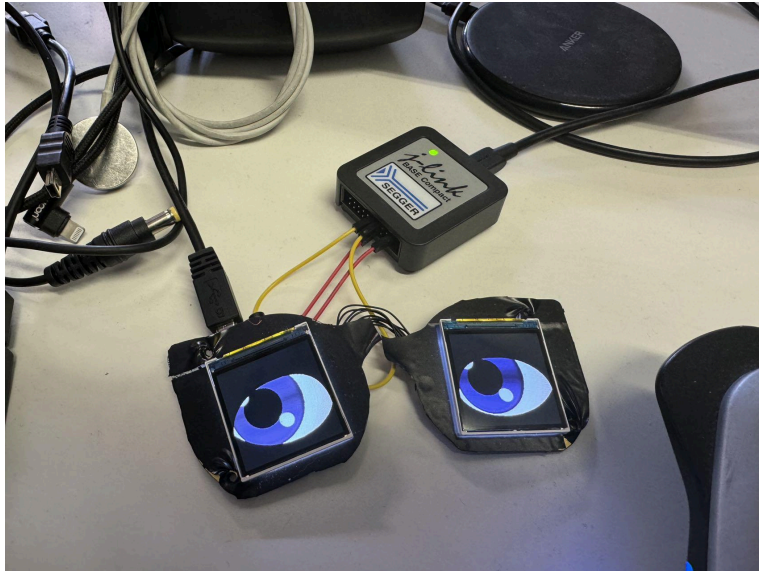




Programming Guide

Dec 6, 2024

Step 1)

Hook the connector of the J-Link programmer to the main eye board. Also, plug in the USB connector on the eye board to supply power to the board.

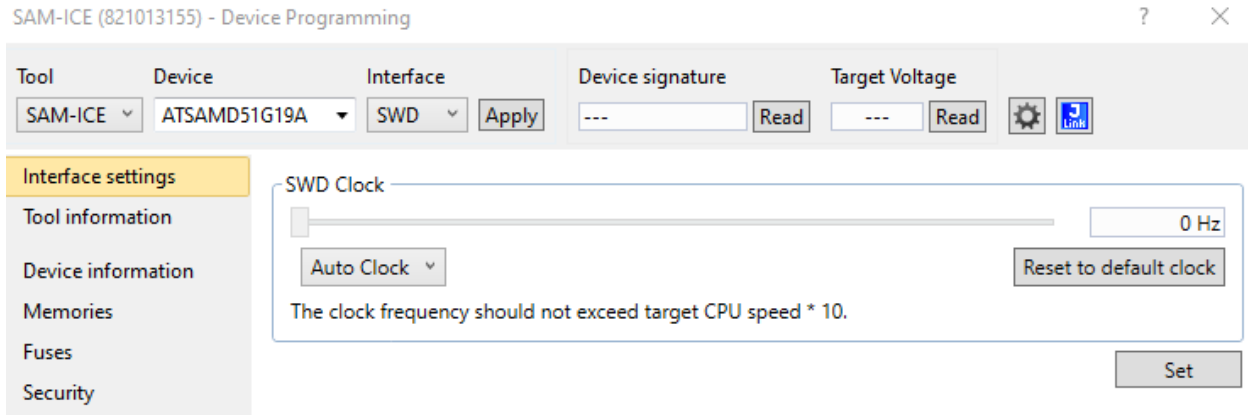
VTref	1	2	NC	
Not used	3	4	GND	3.3v
Not used	5	6	GND	
SWDIO	7	8	GND	SWDIO
SWCLK	9	10	GND	SWCLK
Not used	11	12	GND	GND
SWO	13	14	GND*	
RESET	15	16	GND*	
Not used	17	18	GND*	
5V-Supply	19	20	GND*	

Step 2)

Open [Atmel Studio 7](#). Under the “Tools” menu select “Device Programming”

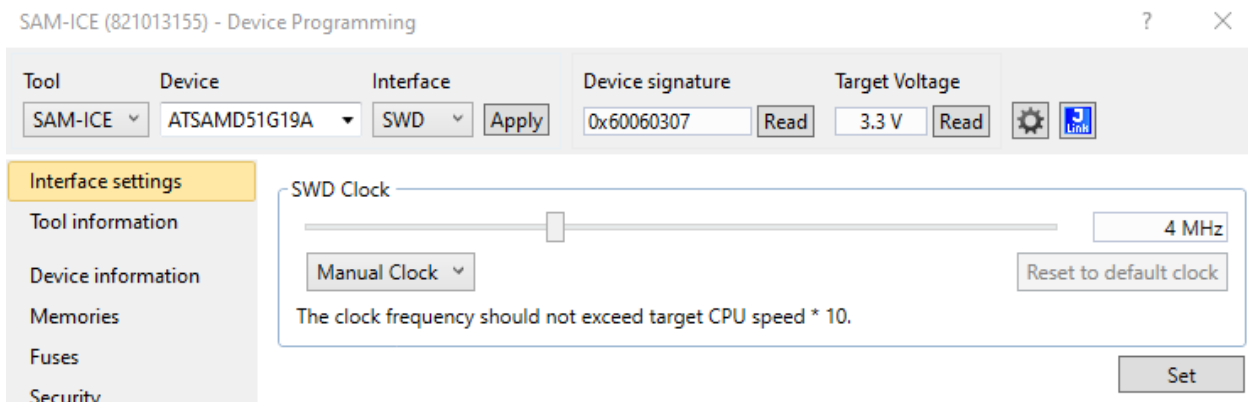
Step 3)

Select the J-Link and under Device select the “ATSAMD51G19A”. Select the “SWD” interface and click Apply.



Step 4)

Now click “Read” under the Device signature



Step 5)

Under “Fuses” make sure they are all set like this. However, set the **USER_WORD_0.NVMCTRL_BOOTPROT** to “0 kbytes” (not shown in the screenshot). Click “Program” to write the fuses:

Tool	Device	Interface	Device signature	Target Voltage
SAM-ICE	ATSAMD51G19A	SWD	0x60060307	3.3 V

Interface settings	Fuse Name	Value
Tool information	SW0_WORD_0.AC_BIAS0	0x01
Device information	SW0_WORD_0.ADC0_BIASCOMP	0x07
Memories	SW0_WORD_0.ADC0_BIASREFBUF	0x04
Fuses	SW0_WORD_0.ADC0_BIASR2R	0x07
Security	SW0_WORD_0.ADC1_BIASCOMP	0x07
	SW0_WORD_0.ADC1_BIASREFBUF	0x04
	SW0_WORD_0.ADC1_BIASR2R	0x07
	SW0_WORD_1.USB_TRANSN	0x09
	SW0_WORD_1.USB_TRANSP	0x19
	SW0_WORD_1.USB_TRIM	0x06
	TEMP_LOG_WORD_0.ROOM_TEMP_VAL_INT	0x1E
	TEMP_LOG_WORD_0.ROOM_TEMP_VAL_DEC	0x00
	TEMP_LOG_WORD_0.HOT_TEMP_VAL_INT	0x68
	TEMP_LOG_WORD_0.HOT_TEMP_VAL_DEC	0x09
	TEMP_LOG_WORD_0.ROOM_INT1V_VAL	0x01
	TEMP_LOG_WORD_1.HOT_INT1V_VAL	0x00
	TEMP_LOG_WORD_1.ROOM_ADC_VAL_PTAT	0x0A94
	TEMP_LOG_WORD_1.HOT_ADC_VAL_PTAT	0x0D26
	TEMP_LOG_WORD_2.ROOM_ADC_VAL_CTAT	0x0B7B
	TEMP_LOG_WORD_2.HOT_ADC_VAL_CTAT	0x095E
	USER_WORD_0.BOD33_DIS	☒
	USER_WORD_0.BOD33USERLEVEL	0x1C
	USER_WORD_0.BOD33_ACTION	The BOD33 generates a reset
	USER_WORD_0.BOD33_HYST	0x02
	USER_WORD_0.NVMCTRL_BOOTPROT	16 kbytes
	USER_WORD_1.NVMCTRL_SEESBLK	0x00
	USER_WORD_1.NVMCTRL_SEEPSZ	0x00
	USER_WORD_1.RAMECC_ECCDIS	☒
	USER_WORD_1.WDT_ENABLE	☐
	USER_WORD_1.WDT_ALWAYS ON	☐
	USER_WORD_1.WDT_PER	16384 clock cycles
	USER_WORD_1.WDT_WINDOW	16384 clock cycles
	USER_WORD_1.WDT_EWOFFSET	16384 clock cycles
	USER_WORD_1.WDT_WEN	☐
	USER_WORD_2.NVMCTRL_REGION_LOCKS	0xFFFFFFFF

Fuse Register	Value
SW0_WORD_0	0x01E7079D
SW0_WORD_1	0x00001B29
TEMP_LOG_WORD_0	0x0196801E
TEMP_LOG_WORD_1	0xD26A9400
TEMP_LOG_WORD_2	0xFF95EB7B
USER_WORD_0	0xF69A9239
USER_WORD_1	0xAECCFF80
USER_WORD_2	0xFFFFFFFF

☒ Auto read
☒ Verify after programming

Copy to clipboard
 Program Verify Read

Reading register TEMP_LOG_WORD_1...OK
 Reading register TEMP_LOG_WORD_2...OK
 Reading register USER_WORD_0...OK
 Reading register USER_WORD_1...OK
 Reading register USER_WORD_2...OK
 Read registers...OK

Read registers...OK

Close

Step 6)

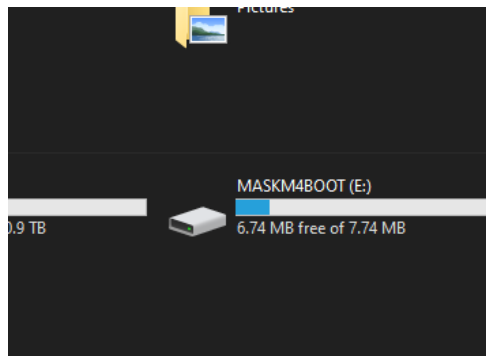
Under “Memories” select the “bootloader-hallowing_mask-v3.16.0.bin” file and click “Program.” When it’s done click “Verify.” The flash should verify correctly.

Step 7)

Go back to “Fuses” and set USER_WORD_0.NVMCTRL_BOOTPROT to “16 kbytes” then click “Program” - It should now look exactly like the screenshot in step 5.

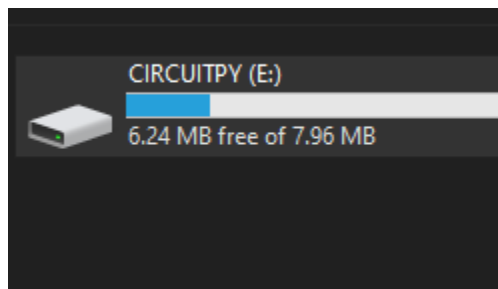
Step 8)

Unplug the J-Link. A drive should mount:



Step 9)

We need to set up the file system on the external flash so first drag the CircuitPython file into that drive “adafruit-circuitpython-monster_m4sk-en_US-9.0.5.uf2”. Once it’s done the MASKM4BOOT drive should unmount and a new drive called CIRCUITPY should mount:

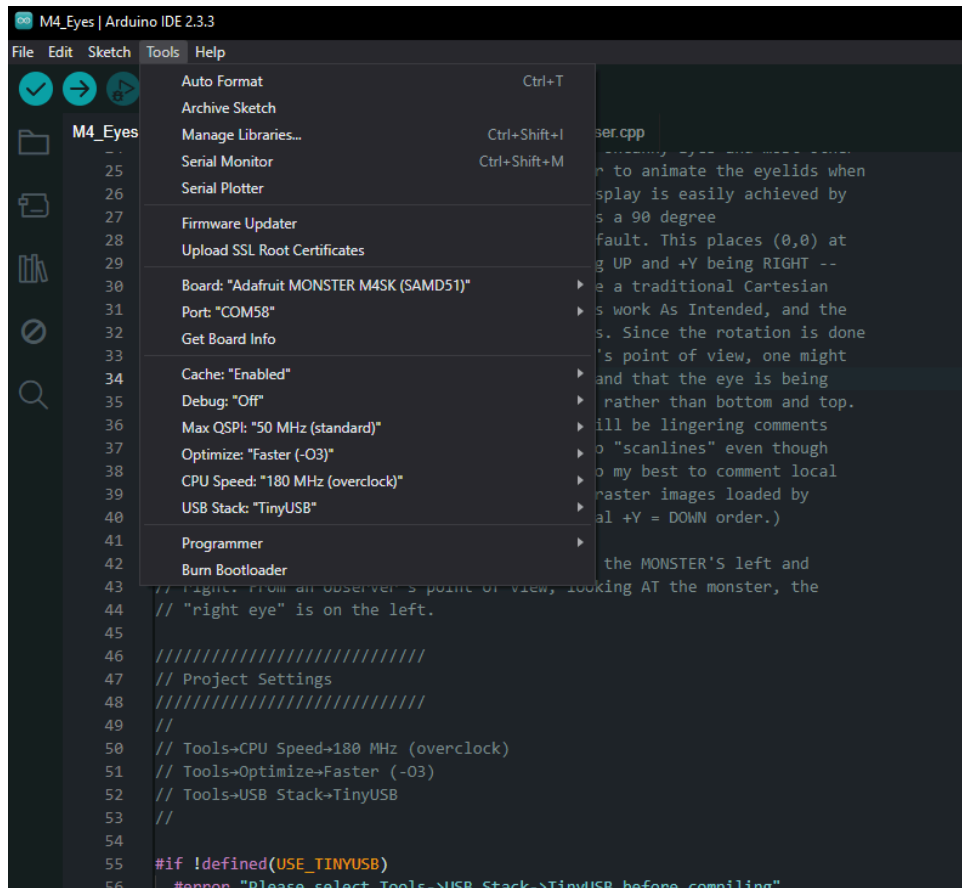


If asked about copying with all the properties, click “yes”.

You should also see some debug information on the eyes LCD screen.

Step 10)

Open the “M4_Eyes” Arduino project, check that you have the correct options set under the tools menu (shown below), and upload it to the Adafruit Monster M4sk Board.



Tools→CPU Speed→180 MHz (overclock)

Tools→Optimize→Faster (-O3)

Tools→USB Stack→TinyUSB

Click on upload.

Step 11)

The CIRCUITPY drive will remount on your computer after the upload. Copy the 3 eye files (two files and one directory) over to the CIRCUITPY drive.

Hackerbot Industries > Software > 2024.11.16 - Eye Board Programming v2 > CIRCUITPY Files					Search CIRCUITPY Files
	Name	Date modified	Type	Size	
★	anime	11/16/2024 8:32 PM	File folder		
★	config.eye	11/14/2024 6:06 PM	EYE File	1 KB	
★	splash.bmp	11/14/2024 6:05 PM	BMP File	169 KB	

Again, check the box for “Do this for all current items” and click “Yes” to confirm you want to copy the files without all the properties.

Step 12)

Done! Hit the reset button on the eye board, and the eyes will be displayed!